Business Calculus (MATH 2350)
Summer 2018
Thursday, 06/14/2018
Quiz 6: Section 12.1, 12.2

Name (Print): ______________________

Score:	/ 110

Section 12.1

1. $\boxed{\text{10 points}}$ If $f(x) = \dfrac{2}{x} - \dfrac{2}{x^2}$, find the intervals on which $f(x)$ is increasing.

2. A manufacturer incurs the following costs in producing x boomerangs in one day, for $0 < x < 120$:

Fixed costs: \$300

unit production cost: \$5 per boomerang

equipment and repairs: $\$0.002x^3$

So, we yield the cost function $C(x) = 300 + 5x + 0.002x^3$.

(a) $\boxed{\text{10 points}}$ What is the average cost $\overline{C}(x)$ per boomerang if x boomerangs are produced in one day?

(b) $\boxed{\text{10 points}}$ Find the intervals on which $\overline{C}(x)$ is increasing, and the intervals on which $\overline{C}(x)$ is decreasing.

3. $\boxed{\text{10 points}}$ Let $f(x) = (x-9)e^{-5x}$. Find the intervals on which $f(x)$ is increasing, the intervals on which $f(x)$ is decreasing, and the local extrema.

4. $\boxed{\text{10 points}}$ If $f(x) = x^3 + 5x - 24$, find the intervals on which $f(x)$ is increasing.

Section 12.2

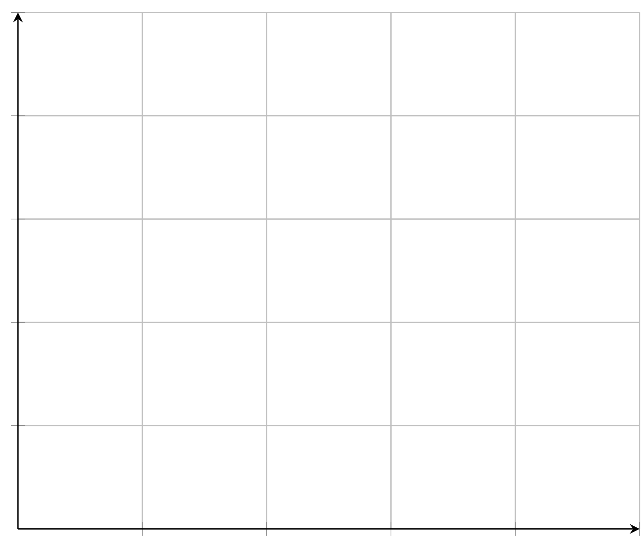
1. $\boxed{\text{10 points}}$ If $y = \dfrac{6}{x} + e^{3x}$, find $\dfrac{d^2y}{dx^2}$.

2. $\boxed{\text{10 points}}$ If $f(x) = 4x^6 + 13x + 13$, find any values of x for which $f''(x) = 0$.

3. $\boxed{\text{20 points}}$ Sketch a graph of function $f(x)$ which is continuous on $(-\infty, \infty)$ and satisfies:

$$f(-2) = 3, \quad f'(0) = 0, \quad f'(-2) = 0, \quad f''(-1) = 0,$$

$$f'(x) < 0 \text{ on } (-2, 0), \text{ and } f'(x) > 0 \text{ on } (0, \infty) \text{ and } (-\infty, -2).$$



4. $\boxed{\text{10 points}}$ If $f(x) = 4x^6 + 13x + 13$, find any values of x for which $f''(x) = 0$.

5. $\boxed{\text{10 points}}$ If $f(x) = x^3 + 18x^2$, find the x and y coordinates of all inflection points.